

Driver Monitoring System (DMS)

Step-by-Step Live Demo Presentation Script

DEMO EXECUTION PLAN

Phase 1: Verified Driver Flow (Registered Driver in DB)

STEP 1 Face Verification & Driver Recognition

- **Action:** Align a registered driver's face in front of the camera on the **Verification Screen**.
- **Mechanism:** FaceAuthEngine matches facial embeddings with local/backend DB embeddings.
- **Result:** Displays "Driver Verified" indicator, plays chime, and retrieves driver profile.

 **WHAT TO SAY DURING DEMO:**
"First, when a registered driver sits in the vehicle, the system performs real-time face authentication using local facial embeddings. The system successfully recognizes the driver and starts the trip session."

STEP 2 Live Monitoring & Telemetry Streaming

- **Action:** App navigates automatically to the **Live Monitoring Panel**.
- **Mechanism:** Processing ~30 FPS camera feed. TelemetryService initiates live frame streaming to dashboard/server.

STEP 3 Triggering & Demonstrating Incident Types

 **Drowsiness Alert**
PERCLOS > 30% or Eye Aspect Ratio (EAR) drops. Audio warning chime plays.

 **Wake Up / Sleeping Alert**
Eyes closed >= 5s OR Head Drop >= 5s. Loud audio "WAKE UP!" alarm + screenshot capture.

 **Phone Usage Detection**
Cell phone detected in hand/ear via ObjectDetectorEngine. Visual alert triggered.

 **Smoking Detection**
Cigarette object classified near mouth/hand. Audio & visual alert triggered.

 **Out of Boundary / Head Distraction**
Yaw angle > 25° for sustained 3s or face moves outside designated camera box. "Driver Distracted" banner displays.

STEP 4 Stop Trip & Return to Verification Screen

- **Action:** Click the "Stop Trip" button on the UI.
- **Result:** Saves trip summary, closes stream, and resets safely to the **Verification Screen**.

 **WHAT TO SAY DURING DEMO:**
"When the trip ends, clicking 'Stop Trip' compiles telemetry metrics, closes the server stream, and returns to the verification screen."

Phase 2: Unverified / Unknown Driver Flow (Person NOT in DB)

STEP 1 Unknown Driver Detection

- **Action:** Unregistered person sits in front of camera.
- **Result:** Face authentication fails to match DB → Tagged as "**Unknown Driver / Unauthorized Access**".

STEP 2 Incident Monitoring & Security Alerting

- **Action:** System auto-starts trip under UNKNOWN driver profile.
- **Triggering Incidents:** Test all 5 incidents (Wake up, Drowsiness, Phone, Smoking, Out of Boundary). All telemetry sent to server tagged with security alert.

STEP 3 Stop Trip Execution

- **Action:** Click "**Stop Trip**" → Ends unknown trip session → Returns to Verification Screen.

Phase 3: Vehicle Speed Trigger Flow (≥ 20 km/h) & Frame Check

STEP 1 Automatic Monitoring Activation (≥ 20 km/h)

- **Action:** Vehicle speed exceeds **20 km/h** (via GPS or accelerometer).
- **Mechanism:** System automatically bypasses waiting state and forces transition to active monitoring.

WHAT TO SAY DURING DEMO:

"If the vehicle begins moving above 20 km/h before manual verification completes, the system automatically forces active monitoring to guarantee continuous driver safety."

STEP 2 Frame Reception Check

- **If Frames Received:** Bounding box renders, EAR/YAW calculations run, incident alerts activate normally.
- **If Frames NOT Received:** System flags camera frame drop / video stream disconnection error.

STEP 3 Stop Trip

- **Action:** Click "**Stop Trip**" → Resets system to Verification Screen.

Summary Reference Table

Scenario	Entry Condition	Incidents Verified	Exit Action
Verified Driver	Face match in Database	Drowsiness, Wake Up (>=5s), Phone, Smoking, Out of Boundary	Stop Trip → Verification Screen
Unknown Driver	Face NOT found in Database	All 5 incidents logged under UNKNOWN profile + Security Flag	Stop Trip → Verification Screen
Speed Trigger	Speed ≥ 20 km/h (GPS/Accel)	Auto-start monitoring + Verify Frame reception & Alerts	Stop Trip → Verification Screen